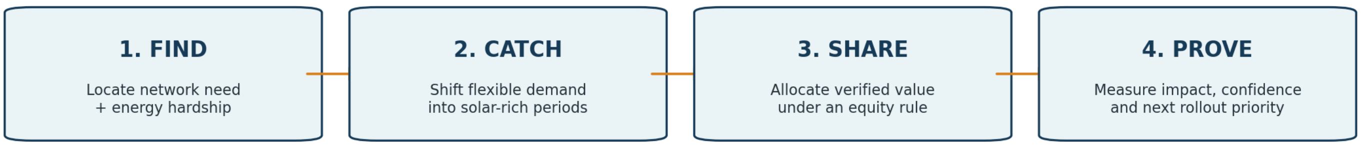
SUNSHINE WALLET

A Local Energy Equity Orchestrator for Wollongong

Energy Equity Challenge - Wollongong City Council



Sunshine Wallet is an equity orchestration layer, not a new electricity grid.

Core proposition

When local solar cannot be fully accommodated, Sunshine Wallet coordinates compatible flexible demand in the right place and time, verifies the network benefit, and allocates the resulting value under an explicit equity policy - so renters and households without rooftop solar can share in the transition.

Working concept and system design brief | 22 August 2026

Executive Summary

Sunshine Wallet addresses a mismatch in the local energy transition: solar generation and community-energy infrastructure are expanding, but the ability to capture their financial benefits is uneven. Renters, apartment residents and energy-hardship households may be unable to install rooftop solar, while solar-dense parts of the distribution network can experience periods where additional export is constrained or difficult to accommodate.

The concept does not pretend electricity can be routed to a chosen person. Instead, it treats flexible demand as a local network resource. In a production trial, the system would identify a network zone where constraint risk and energy hardship overlap; forecast a credible solar-rich window; coordinate compatible loads such as storage hot water, a community battery, EV charging or apartment common loads; estimate what changed versus a no-intervention baseline; and allocate verified program value as a Flexibility Dividend under an Equity Floor.

One-line pitch

We turn otherwise constrained local sunshine into verified flexibility value - and make sure households locked out of rooftop solar share the reward.

What it is	What it is not
An equity orchestration, verification and settlement layer	A peer-to-peer wire that sends Maria's electrons directly to Dinh
A way to coordinate existing controllable energy resources	A claim that all midday solar is free
A council/DNSP/retailer pilot architecture	A claim that Sunshine Wallet itself owns smart meters or community batteries
A transparent governance mechanism for allocating value	An opaque AI that decides who deserves energy

1. The Problem We Are Actually Solving

1.1 Energy access is unequal

Households with suitable roofs and capital can buy rooftop solar, batteries and EV infrastructure and directly reduce their bills. Other households can be structurally excluded even if they live inside the same renewable-rich community. The challenge is therefore not only "how do we generate more renewable energy?" but also "who can access the value created by local generation, storage and flexibility?"

- Renters may not control the roof, switchboard or landlord investment decision.
- Apartment residents may share roofs, embedded networks, common loads or central hot-water systems.
- Low-income households may have the least ability to buy the very assets that unlock the largest energy savings.
- A household can therefore live near abundant solar infrastructure yet receive little direct benefit from it.

1.2 Local solar can create operational constraints

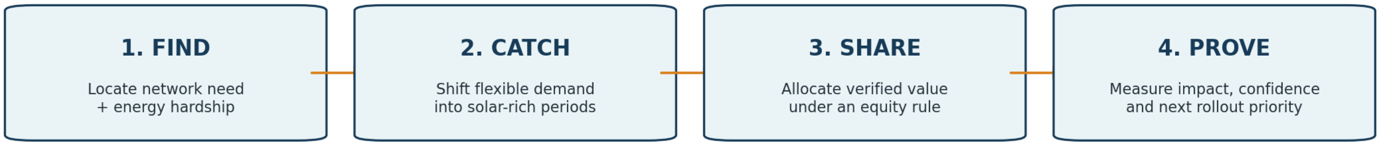
During solar-rich periods, local export can be limited by voltage, thermal capacity, configured export limits or dynamic network settings. These mechanisms are not identical. Sunshine Wallet therefore avoids treating every "missing" kilowatt-hour as the same phenomenon and uses the broader technical language of **otherwise constrained or underutilised local solar capacity**.

Important technical discipline
The emotional phrase "wasted sunshine" is useful in a pitch, but the technical document should distinguish export limits, dynamic export constraints and voltage-related inverter curtailment. The system should only claim a rescued benefit when it can estimate a credible counterfactual.

1.3 The hidden pricing problem

Moving a renter's hot-water heating to midday does not automatically reduce their bill. A flat retail tariff may charge the same cents/kWh at noon and at night. Sunshine Wallet therefore needs an explicit settlement mechanism: verified flexibility must create a bill credit, program credit, network support payment or equivalent retailer-arranged benefit. This is why the concept is a flexibility-and-equity market layer, not merely a scheduling app.

2. Sunshine Wallet: The Solution



Sunshine Wallet is an equity orchestration layer, not a new electricity grid.

2.1 FIND - locate the right intervention zone

The Equity Index combines public and partner data to identify where three things overlap: energy hardship, local renewable opportunity and a network condition that flexible demand could plausibly help. The prototype can score suburbs or synthetic feeder cells; a production deployment should move toward topology-aware network zones.

- Energy equity indicators: income, rental tenure, apartment share, rebate or hardship proxies.
- Renewable opportunity: rooftop solar density and solar forecast.
- Network context: DAPR information, export constraints, available community battery programs and eventually topology/voltage data.
- Confidence: the UI must disclose how precise or approximate each score is.

2.2 CATCH - move useful demand into the solar-rich window

The system forecasts a local period when additional demand is valuable, then chooses only compatible resources that can help that particular constraint. The first realistic category is storage hot water where compatible smart-meter/controlled-load capability exists. Other resources can include community batteries, EV charging and building-level apartment loads.

The central insight

Sunshine Wallet does not move electricity to a disadvantaged person. It moves flexible demand into the time and location where abundant local solar can be better accommodated.

2.3 SHARE - distribute value fairly

There are two distinct channels of benefit, and the system should be explicit about the difference:

Channel	What can be targeted	How fairness works
Controllable appliance	The selected household's demand can be scheduled	Eligibility + rotation + network effectiveness + comfort constraints
Community battery / pooled network value	The economic value, not individual electrons	Verified Flexibility Dividend allocated under a community governance rule

2.4 PROVE - measure whether the intervention worked

After each event, a Verification Engine estimates what would likely have happened without intervention. Credits should be based on verified or conservatively estimated flexibility, not raw consumption. At program level, Council can track kWh accommodated, dollars delivered, participation, confidence and an updated equity score.

3. Why This Is Not a Generic Hackathon Energy App

Generic approach	Sunshine Wallet difference
“Trade solar with your neighbour”	Does not pretend electrons can be routed peer-to-peer; targets demand or financial value instead.
Static hardship map	Uses the map to choose an intervention and then measures what changed.
Battery allocation calculator	Links dispatch to an actual network need and a settlement rule.
“AI tells you to run the dishwasher at noon”	Makes participation automatic where possible; does not shift the burden onto vulnerable households.
Pure network optimisation	Adds an Equity Floor so the best electrical resource does not automatically capture all value.
Opaque AI score	Shows rule weights, confidence and why a household/resource was or was not selected.

3.1 Sunshine Cells: solve the “same feeder” problem

A major hidden failure mode is assuming any consumption on the same feeder relieves any solar constraint. Low-voltage constraints can be local to a transformer, branch or phase. Sunshine Wallet therefore uses the concept of a **Sunshine Cell**: the smallest network zone for which available data can support a meaningful relationship between generation, constraint and flexible demand. In the MVP this can be synthetic or approximate; in production it should use progressively richer network topology and telemetry.

3.2 The Equity Floor

Pure grid optimisation may prefer wealthy households because they are more likely to own EVs, batteries, pool pumps and smart devices. Sunshine Wallet prevents that outcome with a governance constraint:

> **Optimisation principle**
> Maximise verified network value subject to customer comfort, technical compatibility, consent and a minimum Equity Floor - for example, at least 60% of distributable program value must reach qualifying energy-hardship households.

3.3 Equity without device ownership

The most disadvantaged households may have no individually controllable asset. Participation therefore has two paths:

- Flexibility Participant: a compatible household or building provides controllable demand and can receive direct flexibility credits.

- Equity Pool Participant: an eligible household without a compatible device can still receive a share of verified pooled program value, subject to program rules.

This prevents the program from reproducing the same asset-ownership inequality it is intended to address.

4. Why the Concept Is Realistic

4.1 The enabling mechanisms already exist

Existing evidence	Why it matters
Endeavour Energy Off Peak Plus trial used smart meters across 2,500 Albion Park homes to dynamically manage storage hot water and capture more affordable solar energy.	The demand-shifting mechanism is not speculative; controlled hot-water flexibility has already been trialled in the Endeavour network.
Endeavour Energy lists community batteries in Dapto and Warrawong and says participation can include renters and people who do not own their home.	The Illawarra already has relevant community-storage infrastructure and retailer partnership models.
NSW/AEMC smart-meter rollout continues toward 2030.	Smart-meter capability is expanding, but the proposal must not claim every household already has a compatible meter or controllable circuit.
AEMC 2026 pricing review emphasises consumer-led flexibility and pricing reform.	The broader market is moving toward rewarding flexible participation, while also highlighting why settlement and consumer protection matter.

4.2 What Sunshine Wallet adds

- Equity-aware targeting: network optimisation asks what helps the grid; Sunshine Wallet also asks who has been excluded from local renewable benefits.
- Topology-aware matching: flexible demand is matched to the location of the constraint, not merely the suburb name.
- Verification: the platform estimates additional benefit relative to a baseline before writing credits.
- Governance: the allocation policy is explicit, adjustable and bounded by an Equity Floor.
- Explainability: residents can see why they received a credit without seeing electrical-engineering complexity.
- Privacy-by-design: the platform should receive an eligibility token rather than detailed welfare or income data wherever possible.

4.3 What must remain a pilot assumption

Do not claim	Defensible replacement
“Every smart meter has the required hot-water contactor.”	“Phase 1 enrols households with compatible smart-meter/controlled-load infrastructure.”
“Zero new hardware.”	“No bespoke household hardware is required for compatible participants; some sites may require metering or electrical upgrades.”
“Nobody pays because the fuel is waste.”	“The energy opportunity may be otherwise underutilised, but coordination, metering, battery operation and settlement still have costs.”
“Solar owners lose nothing.”	“The program is designed not to reduce normal self-

	consumption and only acts within agreed network/market conditions.”
“Same feeder means the energy is rescued.”	“The intervention must match the relevant network zone and be verified against a counterfactual.”

5. Application Experience

5.1 Resident app - hardship/renter journey

1. Welcome: “Turn local sunshine into lower energy bills. You do not need solar panels.”
2. Address/program check: identify whether the household belongs to an active Sunshine Cell or pilot area.
3. Eligibility: verify through an authorised party where possible; Sunshine Wallet stores only the minimum eligibility result needed.
4. Device capability: identify compatible storage hot water, EV, battery or building-level load. If none exists, offer Equity Pool participation.
5. Consent: plain-English explanation, customer-comfort guarantee, pause/opt-out.
6. Wallet home: monthly dollar benefit first; kWh and technical details secondary.
7. Event transparency: “Your hot water is scheduled 12:20-1:05pm. No action needed. Pause today.”
8. Explainability: show the event, verified flexibility, credit rule and confidence in simple language.

Resident design rule
The resident product should feel simple and reassuring. The user should not need to understand voltage, DAPR, phases or battery dispatch to benefit.

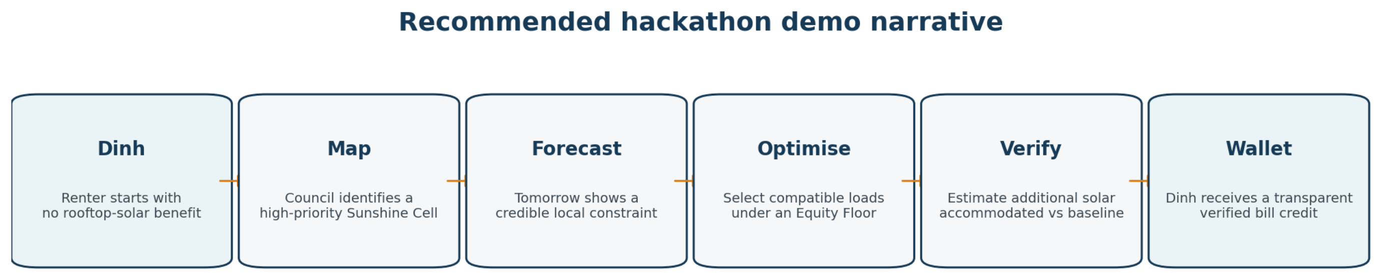
5.2 Solar-owner journey

- Show generation, exported energy and additional local solar estimated to have been accommodated during qualifying events.
- Show any contributor reward separately from the equity allocation.
- Avoid moralising language: participation is framed as helping local network efficiency and community access, not “giving away” owned electricity.

5.3 Council / operator console

- Equity Opportunity Map: priority areas scored by network need × equity need × flexibility availability × confidence.
- Sunshine Cell detail: forecast solar, baseline demand, constraint window, available resources and data confidence.
- Daily Scheduler: proposed dispatch plan and reasons for exclusions.
- Governance Panel: Equity Floor, needs weighting, contributor reward, community reserve and live cohort outcomes.
- Verification: baseline vs actual; estimated additional solar accommodated; confidence range.
- Impact Reporting: kWh, dollars, households, opt-outs, fairness share, curtailment/constraint indicators and rollout ranking.

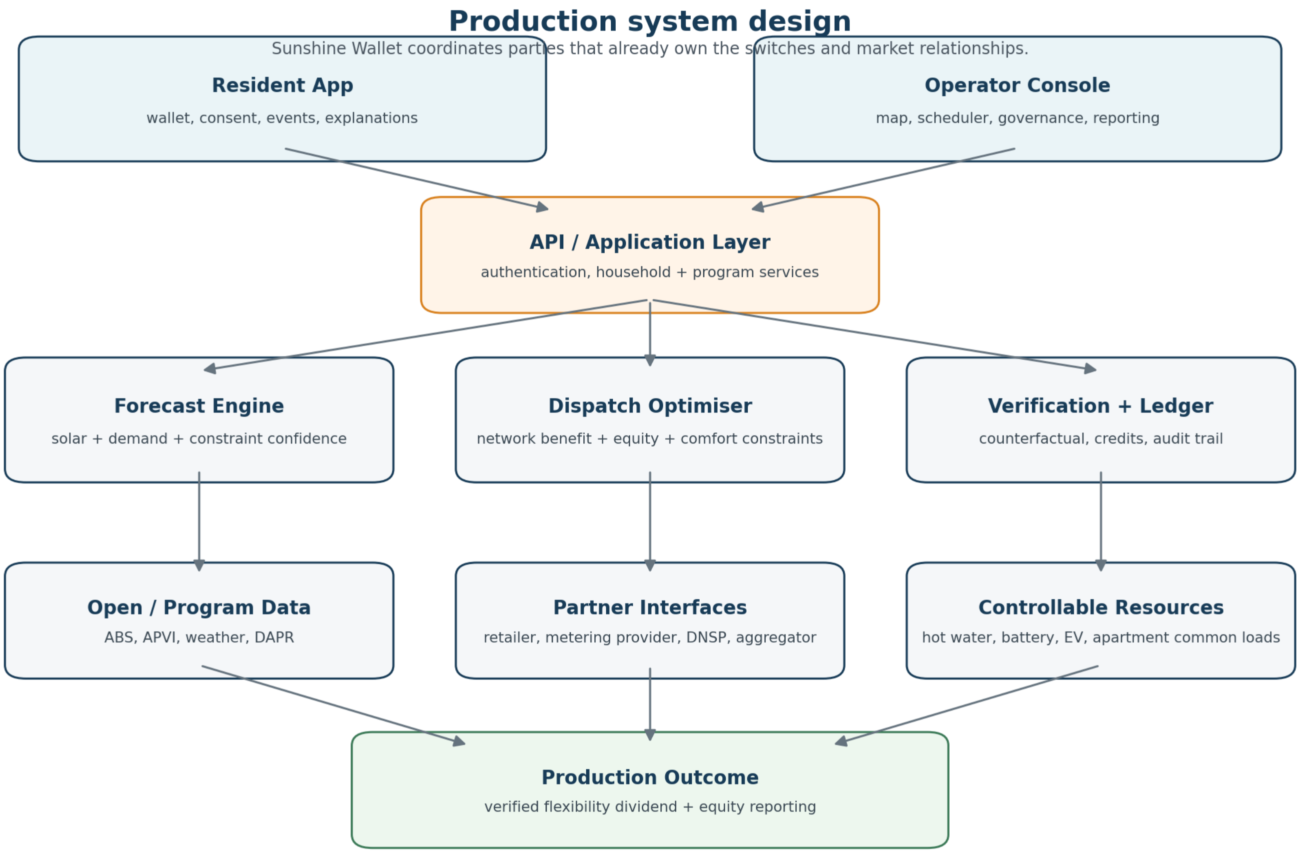
6. Recommended Hackathon Demo Flow



The audience sees one complete cause-and-effect story rather than a tour of unrelated screens.

Beat	What the judges see	Why it matters
1. Dinh	Renter in Dapto starts with no rooftop-solar benefit.	Makes the challenge human immediately.
2. Council map	A high-priority Sunshine Cell is selected because hardship and network opportunity overlap.	Shows data is used to decide, not decorate.
3. Forecast	A solar-rich constraint window appears with a confidence score.	Introduces the real technical problem and uncertainty.
4. Optimise	Hot-water groups and a community battery are selected; an incompatible resource is rejected.	Shows the engine can say no and respects topology/comfort.
5. Before/after	Constraint risk falls; estimated additional solar accommodated is shown versus baseline.	Provides a visible technical payoff.
6. Governance	Move the Equity Floor and watch cohort outcomes change; the UI refuses an invalid policy.	Makes fairness operational rather than rhetorical.
7. Wallet	Return to Dinh: transparent monthly Flexibility Dividend with explanation.	Closes the loop emotionally and economically.

7. System Design



7.1 Major services

Service	Responsibility
Identity & Consent	Users, households, roles, program enrolment, consent and pause/opt-out.
Sunshine Cell Service	Maps addresses/resources to the appropriate network zone and stores equity/confidence metadata.
Forecast Engine	Predicts solar, baseline demand and a probable network need window.
Resource Registry	Tracks controllable loads, compatibility, power limits, state of charge and comfort constraints.
Dispatch Optimiser	Selects resources to meet network need while respecting equity, rotation, consent and device constraints.
Verification Engine	Estimates baseline vs actual and calculates verified flexibility with confidence.
Settlement / Ledger	Applies governance rules and writes auditable

	household/program credits.
Reporting & Governance	Council dashboards, policy simulation, audit trail and rollout ranking.

7.2 Core optimisation logic

Simplified objective

Maximise: Network Benefit + Equity Benefit. Subject to: correct Sunshine Cell, device availability, customer comfort, consent, battery limits, rotation fairness, Equity Floor and a minimum confidence threshold.

For the hackathon, a weighted score is sufficient and explainable, for example: 35% network effectiveness + 30% equity need + 20% controllability + 15% fairness rotation. The team should present these as prototype weights, not scientifically proven universal weights.

7.3 Verification logic

The key formula is conceptual rather than falsely precise:

Verified flexibility

Observed response during the event - estimated counterfactual response without the event = verified/estimated flexibility. Credit only the defensible portion, and show a confidence score.

7.4 Minimum data model

Entity	Important fields
Household	id, Sunshine Cell, eligibility token, household type, consent, retailer/program
Device	type, household/building, controllable, power, availability, comfort/safety constraints
Sunshine Cell	suburb, topology identifier, capacity/constraint metadata, equity score, confidence
Energy Interval	timestamp, solar forecast, baseline demand, observed demand, constraint estimate
Flex Event	cell, start/end, expected network need, confidence, status
Dispatch	event, resource, scheduled start/end, expected and verified contribution
Credit	household, event, amount, reason, policy version, audit metadata

8. Hidden Failure Modes and How We Design Around Them

Failure mode	Why it can break the concept	Design response
Flat retail tariff	Shifted hot water may cost the household the same.	Explicit Flexibility Dividend / retailer settlement layer.
Wrong network location	Demand elsewhere on a feeder may not relieve a transformer/phase constraint.	Sunshine Cells + topology-aware matching + confidence.
No controllable asset	The poorest household may own the least flexible equipment.	Equity Pool participation plus building-level resources.
Apartment complexity	Embedded networks or central services break household-level assumptions.	Support common hot water, common-area HVAC/EV charging or financial-only participation.
Cold-shower risk	Aggressive optimisation harms customer comfort.	Temperature/service constraints, minimum reserve, override and opt-out.
Gaming	Rewarding raw noon consumption encourages waste.	Reward verified shifted flexibility against a baseline, not consumption.
Clouds / forecast error	Expected surplus disappears.	Confidence thresholds, re-optimisation and “no rescue event” state.
Battery is not ours	Council software cannot simply command network assets.	Partner interface to retailer/aggregator/DNSP; Sunshine Wallet coordinates rather than owns switches.
Sensitive eligibility data	Collecting welfare/income detail increases privacy risk.	Eligibility-without-exposure: store a minimum eligibility token where possible.
Fake precision	Open data may not prove feeder-level curtailment.	Label simulated/estimated values, show confidence and production data requirements.

8.1 The system must be allowed to do nothing

A production-quality decision engine should sometimes return “No event today.” If forecast confidence is low, the network need is insignificant, the resource is in the wrong Sunshine Cell, or customer comfort would be compromised, Sunshine Wallet should refuse to dispatch. This is a credibility feature, not a weakness.

9. Weekend MVP: What to Build for Real

Build now	Mock / simulate	Production path only
Resident renter wallet	Eligibility response	Live retailer billing integration
Solar-owner impact view	Smart-meter switching	Metering provider control APIs
5-6 area equity map	Sunshine Cell topology	DNSP live voltage/topology feeds
One synthetic Dapto/Warrawong cell	24h solar + demand + constraint data	Real feeder/transformer calibration
Before/after dispatch chart	Hot-water and battery dispatch	Battery aggregator integration
Fairness governance controls	Flexibility Dividend values	Formal tariff / settlement product
Verification + confidence calculation	Counterfactual baseline	Regulatory-grade M&V methodology

9.1 MVP technical stack

- Frontend: React / Next.js with two role-based experiences: Resident and Operator.
- Backend: Node.js / Express or NestJS REST API.
- Database: PostgreSQL/Supabase for users, households, cells, events, dispatches and credits.
- Simulation engine: interval-based solar, baseline demand, capacity, battery state-of-charge and device availability.
- Mapping: Mapbox/Leaflet or a lightweight SVG/GeoJSON map; real suburb statistics can be combined with clearly labelled synthetic network-cell data.
- No fragile live integrations are required for the demo; partner calls should be represented through clean adapter interfaces and mocked responses.

9.2 What not to spend the weekend building

- Real authentication complexity beyond a clean demo login.
- A full electricity retailer billing engine.
- Actual smart-meter control.
- Machine learning merely to say “we used AI.” A transparent forecast and optimiser is more defensible.
- Dozens of screens. One complete cause-and-effect journey is stronger.

10. Fit Against the Judging Criteria

Criterion	How Sunshine Wallet should earn the score
Fairness & equity impact	Targets households structurally excluded from rooftop solar; prevents asset-rich households from capturing all value through an Equity Floor; includes people without controllable assets via the Equity Pool.
Feasibility & grounding	Builds on real smart-meter hot-water trials, community batteries and emerging flexible-energy market design; acknowledges retailer, metering, DNSP and topology constraints rather than hand-waving them.
Design & technical execution	Demonstrates a working interval simulation, optimisation, rejection states, confidence, verification, governance maths and auditable wallet credits.
Presentation & pitch	Starts with a human problem, makes the electricity physics understandable, shows a visually satisfying before/after event, then closes the loop in Dinh's wallet.

10.1 The strongest differentiation sentence

Positioning

Existing network programs ask: “Which flexible load best solves the constraint?” Sunshine Wallet adds: “Which intervention solves the constraint, who has historically been excluded from the benefit, how do we verify the result, and how should the value be shared?”

11. Pitch Language and Defensible Claims

11.1 15-second explanation

15 seconds

Sunshine Wallet finds neighbourhoods where local solar opportunity and energy hardship overlap. When the network needs help absorbing solar, it coordinates compatible flexible loads, verifies the result, and returns the value as transparent bill credits - with an Equity Floor so renters and hardship households are guaranteed a share.

11.2 One-line version

One line

We turn otherwise constrained lunchtime sunshine into lower effective energy costs for people who cannot own solar.

11.3 Technical judge version

Technical version

Sunshine Wallet is a topology-aware local flexibility orchestration, measurement-and-verification, and equity-settlement layer. It coordinates resources already controlled by retailers, metering providers, DNSPs and aggregators rather than claiming to own those assets itself.

11.4 Reality-check panel for the demo

We do not claim...	We handle it by...
Electricity can be routed to individuals	Targeting controllable consumption or allocating financial value.
Same suburb = same electrical constraint	Using Sunshine Cells and data confidence.
Shifted electricity is automatically cheaper	Using an explicit Flexibility Dividend settlement mechanism.
Every renter has compatible equipment	Offering multiple participation modes and an Equity Pool.
Battery access is automatic	Using retailer/aggregator/DNSP partner interfaces.
Every estimated kWh is definitely “rescued”	Counterfactual verification and confidence bounds.

12. Pilot and Scale Roadmap

Stage	Scope	Success evidence
Hackathon MVP	One synthetic Sunshine Cell + 5-6-area equity map + wallet + governance simulator	Working end-to-end logic and transparent assumptions.
Partner discovery	Council + Endeavour + retailer/aggregator + metering provider	Confirm candidate network zone, controllable assets, settlement path and data access.
Small opt-in pilot	One constrained zone, compatible hot-water households and/or community battery	Customer comfort preserved; measurable response; verified value; equitable distribution.
Evaluation	Compare technical and equity outcomes against baseline	kWh accommodated, network metric improvement, bill credits, participation and fairness share.
Rollout	Rank the next Sunshine Cells	Repeatable data pipeline, policy governance and partner operating model.

13. Data and Evidence Strategy

Data / source	MVP use	Production caution
ABS / profile.id	Income, tenure, apartment and demographic equity indicators	Use aggregated data; do not infer individual hardship from suburb averages.
APVI solar data	Solar uptake / generation opportunity proxies	Suburb-level data does not prove feeder-level constraint.
Endeavour DAPR / connection information	Network planning context and constraints	May not expose live LV topology/phase detail needed for precise dispatch.
Weather / solar forecast	Generate daily solar profile	Forecast uncertainty must be shown.
NSW social programs	Design eligibility pathways / equity indicators	Access to individual eligibility must follow authorised processes and privacy rules.
Interval meter / voltage data	Production verification and baseline	Requires customer consent, authorised access and appropriate market arrangements.

Data honesty rule

If a value is simulated, label it simulated. If a network location is approximate, label it approximate. Confidence disclosure is part of the product.

14. Selected Evidence and References

- **Endeavour Energy - Smart hot water storage set to soak up solar (Off Peak Plus, Albion Park, 2,500 homes).**
  https://www.endeavourenergy.com.au/news/media-releases/smart-hot-water-storage-set-to-soak-up-solar
- **Endeavour Energy - Community batteries (including Dapto and Warrawong; retailer participation information).**
  https://www.endeavourenergy.com.au/for-your-home/solar-and-battery-options/community-batteries
- **Endeavour Energy - 2025 community battery program announcement for Illawarra and other regions.**
  https://www.endeavourenergy.com.au/About-us/Newsroom/world-leading-ai-technology-to-unlock-electricity-bill-savings-and
- **AEMC - The Pricing Review: Electricity pricing for a consumer-driven future, final report 18 June 2026.**
  https://www.aemc.gov.au/market-reviews-advice/pricing-review-electricity-pricing-consumer-driven-future
- **NSW Government - What the national smart meter rollout means for you (rollout continuing to 2030).**
  https://www.energy.nsw.gov.au/media/15741
- **NSW Consumer Energy Strategy - smart meters, demand response and consumer participation.**
  https://www.energy.nsw.gov.au/sites/default/files/2024-09/NSW_Consumer_Energy_Strategy_2024.pdf

15. Team Build Checklist

- ☐ Lock the product definition: equity orchestration + verification + settlement, not a generic solar-sharing marketplace.
- ☐ Choose one hero Sunshine Cell and make every screen tell the same story.
- ☐ Define prototype data as real, estimated or simulated - never blur the categories.
- ☐ Implement a “no event” state and at least one resource rejection reason.
- ☐ Implement an Equity Floor that can block an unfair governance setting.
- ☐ Implement a counterfactual/verification calculation with a confidence score.
- ☐ Keep resident UI simple: dollars, reason, control, opt-out.
- ☐ Keep operator UI technical: map, forecast, dispatch, verification, policy and impact.
- ☐ Use a partner-adapter architecture instead of pretending Sunshine Wallet directly controls every asset.
- ☐ Prepare answers for tariff/settlement, transformer/phase locality, apartment/embedded-network cases, privacy, gaming and customer comfort.
- ☐ End the pitch back in Dinh's wallet so the technical system resolves into a human outcome.

Final design principle

The strongest version of Sunshine Wallet is not the one with the most features. It is the one that understands where the real electricity system makes equity difficult - and makes those constraints visible, measurable and governable.